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The relative velocity of B with respect to A, $V_{BA} = 80$ km h^{-1} due west

To a passenger in train A, the train B will appear to move westwards with a velocity of 80 km h^{-1}

EXAMPLE 2.28

Consider two trains A and B moving along parallel tracks with the same velocity in the same direction. Let the velocity of each train be 50 km h^{-1} due east. Calculate the relative velocities of the trains.

Solution

Relative velocity of B with respect to A,

$v_{BA} = v_B - v_A$

$= 50 \text{ km h}^{-1} + (-50) \text{ km h}^{-1}$

$= 0 \text{ km h}^{-1}$

Similarly, relative velocity of A with respect to B i.e., v_{AB} is also zero.

Thus each train will appear to be at rest with respect to the other.

EXAMPLE 2.29

How long will a boy sitting near the window of a train travelling at 36 km h^{-1} see a train passing by in the opposite direction with a speed of 18 km h^{-1}. The length of the slow-moving train is 90 m.

Solution

The relative velocity of the slow-moving train with respect to the boy is = (36 + 18) km h^{-1} = 54 km h^{-1} = = $54 \times \frac{5}{18}$ m s^{-1} = 15 m s^{-1}

Since the boy will watch the full length of the other train, to find the time taken to watch the full train:

We have, $15 = \frac{90}{t}$

or $t = \frac{90}{15} = 6 \text{ s}$

EXAMPLE 2.30

A swimmer's speed in the direction of flow of a river is 12 km h^{-1}. Against the direction of flow of the river the swimmer's speed is 6 km h^{-1}. Calculate the swimmer's speed in still water and the velocity of the river flow.

Solution

Let v_s and v_r, represent the velocities of the swimmer and river respectively with respect to ground.

$$v_s + v_r = 12 \quad (1)$$

and

$$v_s - v_r = 6 \quad (2)$$

Adding the both equations (1) and (2)

$2v_s = 12 + 6 = 18$ km h^{-1} or $v_s = 9$ km h^{-1}

From Equation (1),

$9 + v_r = 12$ or

$v_r = 3$ km h^{-1}

When the river flow and swimmer move in the same direction, the net velocity of swimmer is 12 km h^{-1}.

Accelerated Motion

During non-uniform motion of an object, the velocity of the object changes from instant to instant i.e., the velocity of the object is no more constant but changes